

Lecture 4

Understanding op-amp models

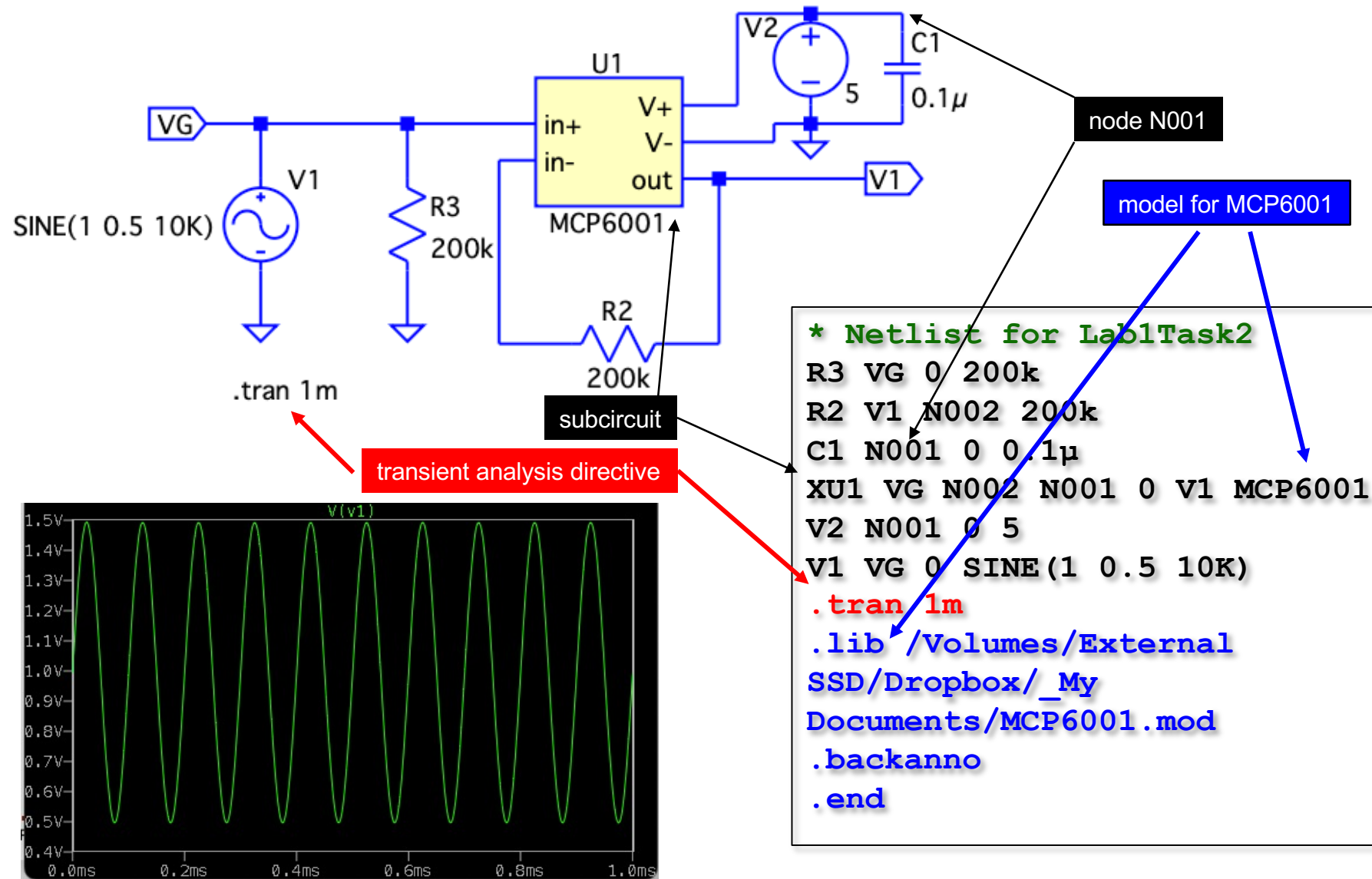
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What is SPICE Simulator?

- ❖ Simulation Program with Integrated Circuit Emphasis
 - Developed by Larry Nagel at Berkeley in 1970's (under Prof Don Pederson).
 - Many commercial and freeware versions available.
 - Most significant CAD tool that drove the advancements in microelectronics.
- ❖ Linear Technology SPICE (LTspice)
 - Developed by Mike Engelhardt in early 2000s when he worked for Linear Technology (later bought by Analog Devices).
 - Added schematic capture and results plotting.
 - Engelhardt now has his own company marketing Qspice – not free.
- ❖ Input to SPICE is a SPICE netlist – multiple lines of text that describe:
 - Components and voltage/current sources in a circuit.
 - How the components and sources are connected to each other.
 - Model of components.
 - Type of analysis to be performed by the simulator

Example of a Spice netlist



SPICE Cheat Sheet

Letter	Component
R	Resistor
C	Capacitor
L	Inductor
V	independent voltage source
I	Independent current source
M	MOSFET
D	Diode
Q	Bipolar transistor
X	Subcircuit
E	Voltage-controlled voltage source
G	voltage-controlled current source

Letter	Unit	Multiplier
T, t	tera	10 E+12
G, g	giga	10 E+9
MEG, meg	mega	10 E+6
K, k	kilo	10 E+3
M, m	milli	10 E-3
U, u	micro	10 E-6
N, n	nano	10 E-9
P, p	pico	10 E-12

Directive	Action
.op	DC operating point analysis
.ac	Small signal AC analysis
.tran	Transient analysis
.backanno	Annotate current back to ports
.include	Include another file
.lib	Include a library
.end	End of netlist
.ends	End of subcircuit
.ic	Set initial condition

* Netlist for Lab1Task2

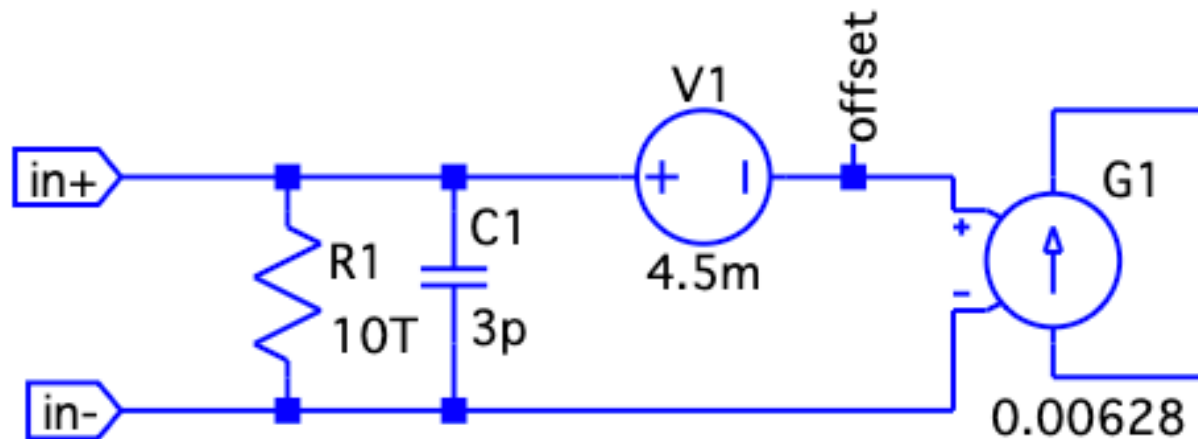
```

R3 VG 0 200k
R2 V1 N002 200k
C1 N001 0 0.1μ
XU1 VG N002 N001 0 V1 MCP6001
V2 N001 0 5
V1 VG 0 SINE(1 0.5 10K)
.tran 1m
.lib /Volumes/External
SSD/Dropbox/_My Documents/MCP6001.mod
.backanno
.end

```

Model input stage of MCP6001

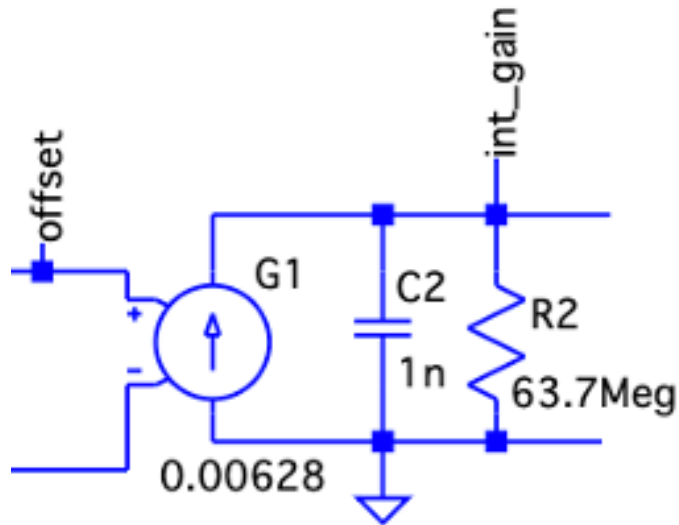
Differential Input Impedance	Z_{DIFF}	—	$10^{13} 3$	—	ΩpF
Input Offset Voltage	V_{OS}	-4.5	—	+4.5	mV



```

R1 in+ in- 10T
C1 in+ in- 3p
V1 in+ offset 4.5m
G1 0 int_gain offset in- 0.00628
    
```

Model gain vs frequency & slew rate of MCP6001

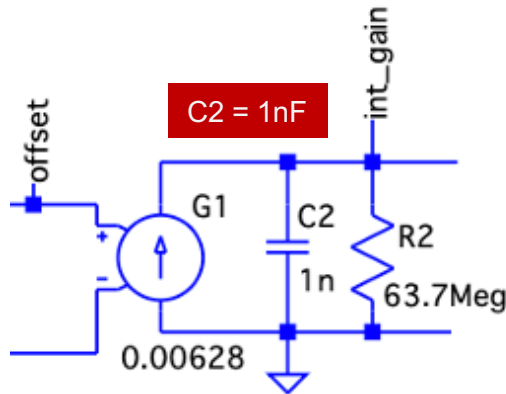


```
...
G1 0 int_gain offset in- 0.00628
R2 int_gain 0 63.7Meg
C2 int_gain 0 1n
```

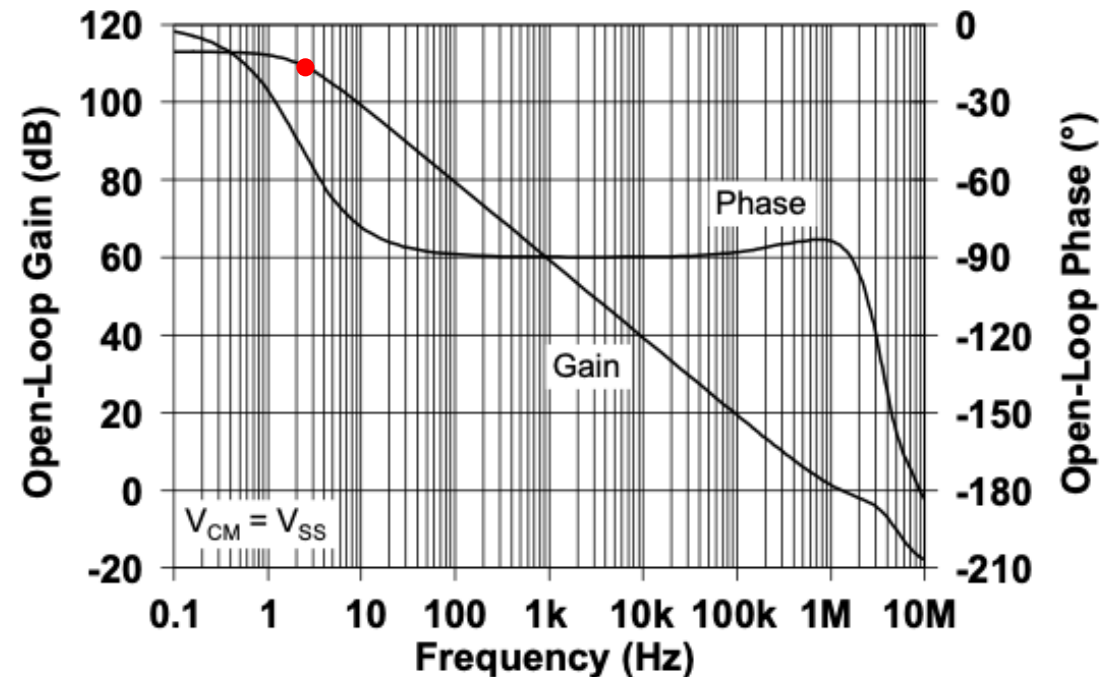
- ❖ Assume compensation capacitor C2 is 1nF

AC Response						
Gain Bandwidth Product	GBWP	—	1.0	—	MHz	G = +1 V/V
Phase Margin	PM	—	90	—	°	
Slew Rate	SR	—	0.6	—	V/μs	
DC Open-Loop Gain (Large Signal)	A _{OL}	88	112	—	dB	

Model open-loop gain vs frequency of MCP6001



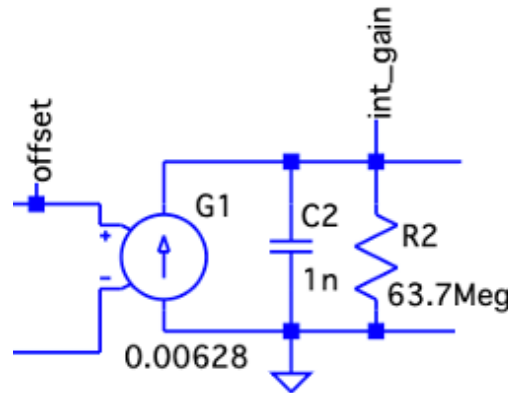
```
...
G1 0 int_gain offset in- 0.00628
R2 int_gain 0 63.7Meg
C2 int_gain 0 1n
```



- ❖ Dominant pole frequency $f_p = \frac{GBP}{A_{OL}} = \frac{10^6}{400000} = 2.5\text{Hz} = 1/2\pi R_2 C_2$.
- ❖ Therefore $R_2 = 63.7 \times 10^6 \Omega$
- ❖ Calculate g_m for G1:

$$g_m \times R_2 = A_{OL}, \quad \text{hence } g_m = \frac{400,000}{63.7} \times 10^{-6} = 0.00628$$

Model Slew rate limit



AC Response

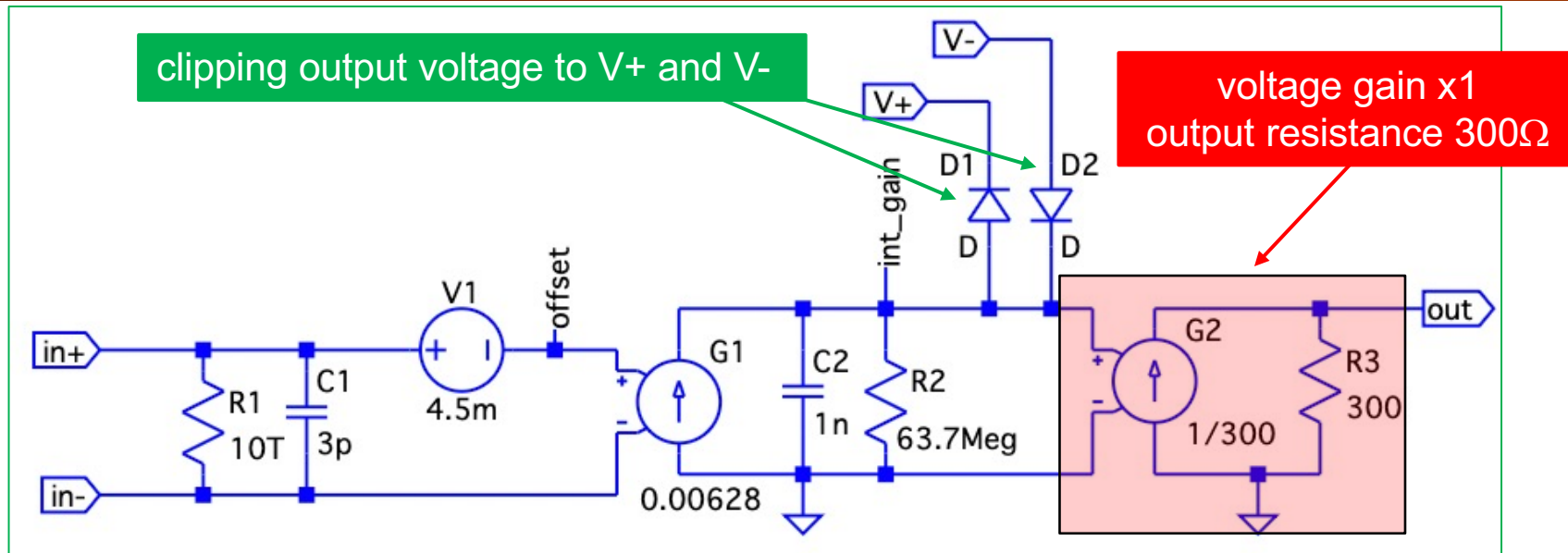
Gain Bandwidth Product	GBWP	—	1.0	—	MHz	
Phase Margin	PM	—	90	—	°	G = +1 V/V
Slew Rate	SR	—	0.6	—	V/μs	

❖ Model the slew rate limit of 0.6V/μs:

$$SR = \max \frac{dV_{C2}}{dt} = \max(\text{current of } G1) / C2, \quad \text{therefore } \max(\text{current}) = SR \times C2 = 0.6 \text{mA}$$

```
G1 0 int_gain value={limit(0.00628*V(offset,in-),0.6m, -0.6m) }
R2 int_gain 0 63.7Meg
C2 int_gain 0 1n
```

Model the output stage of MCP6001



Maximum Output Voltage Swing	V_{OL}, V_{OH}	$V_{SS} + 25$	—	$V_{DD} - 25$	mV	$V_{DD} = 5.5V$, 0.5V input overdrive
Output Short-Circuit Current	I_{SC}	—	± 6	—	mA	$V_{DD} = 1.8V$
		—	± 23	—	mA	$V_{DD} = 5.5V$

* output stage - current limit to +/- 20mA, $R_{OUT} = 300 \text{ ohm}$

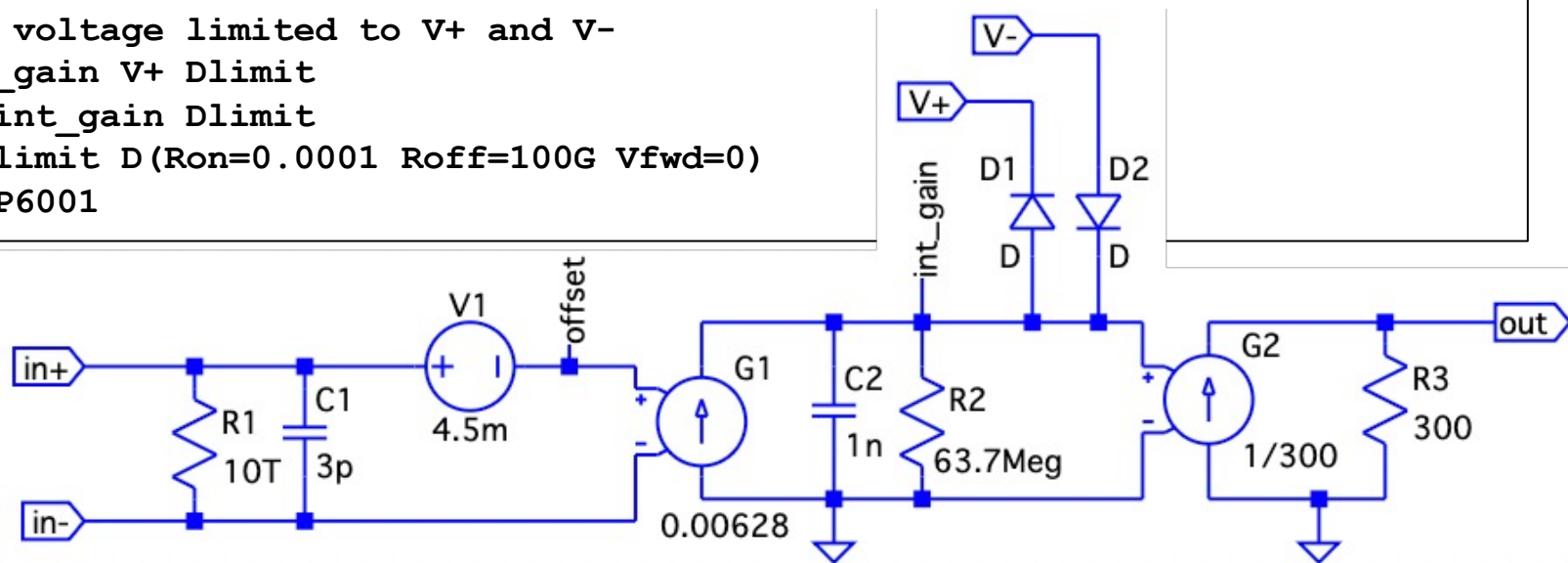
```
G2 0 out value={limit(V(int_gain, 0)/300, 20m, -20m)}
R3 out 0 300
```

* output voltage limited to V+ and V-

```
D1 int_gain V+ Dlimit
D2 V- int_gain Dlimit
.model Dlimit D(Ron=0.0001 Roff=100G Vfwd=0)
```

The complete model of MCP6001

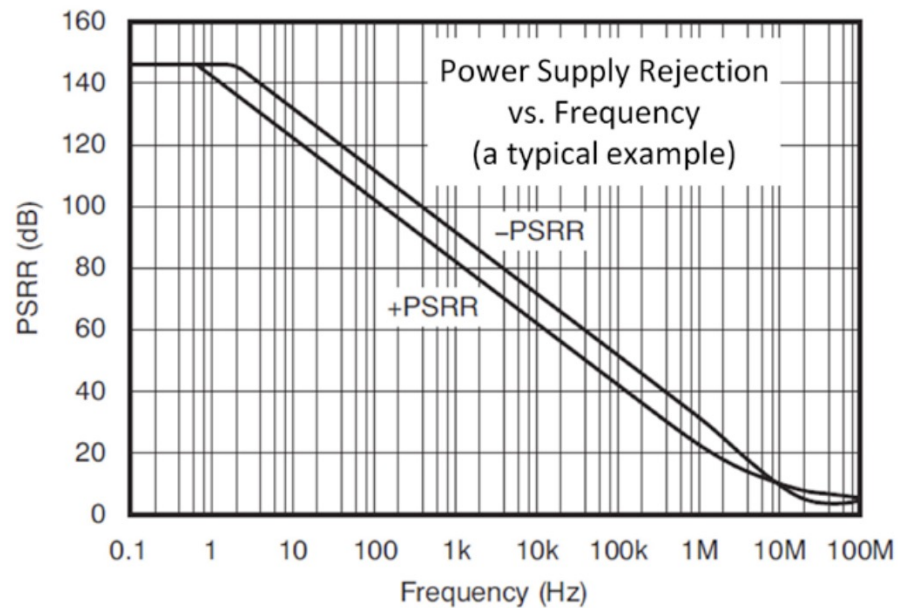
```
.subckt MCP6001 in+ in- V+ V- out
* input stage - RIN = 10T, CIN = 3p, Voffset = 4.5m
  R1 in+ in- 10T
  C1 in+ in- 3p
  Voffset in+ offset dc 4.5m
* gain stage - R2 = {AOL/(6.28*GBP*CPOLE)}, AOL = 400k, GBP = 1Meg, CPOLE = 1n
* gm = 6.28*GBP*CPOLE, current limit IMAX = +/- 0.6mA
  G1 0 int_gain value={limit(0.00628*V(offset,in-),0.6m, -0.6m)}
  R2 int_gain 0 63.7Meg
  C2 int_gain 0 1n
* output stage - current limit to +/- 20mA, ROUT = 300 ohm
  G2 0 out value={limit(V(int_gain, 0)/300, 20m, -20m)}
  R3 out 0 300
* output voltage limited to V+ and V-
  D1 int_gain V+ Dlimit
  D2 V- int_gain Dlimit
.model Dlimit D(Ron=0.0001 Roff=100G Vfwd=0)
.ends MCP6001
```



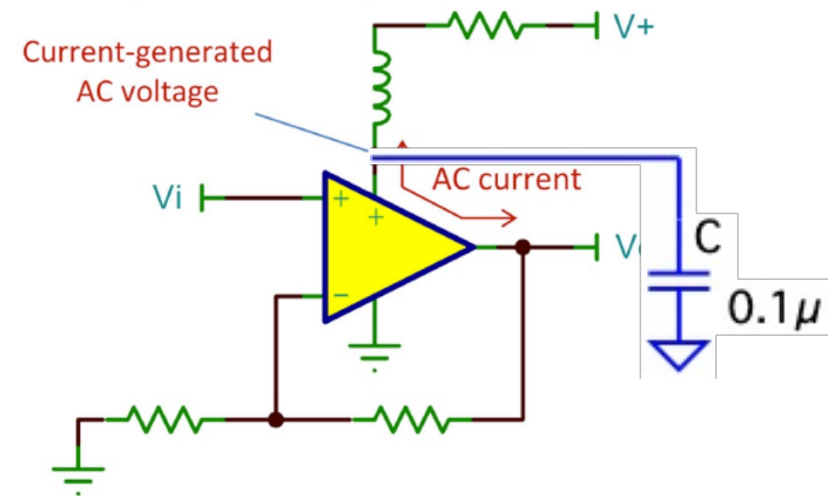
What specifications are NOT modelled?

- ❖ Common mode input impedances
- ❖ Common mode rejection ratio
- ❖ Power supply rejection ratio
- ❖ Common mode input voltage clipping
- ❖ Temperature effect on input bias current and offset voltage
- ❖ Noise characteristics of the op-amp
- ❖ Quiescent current of the op-amp

Power supply bypass (decoupling) capacitor



Power supply without proper bypass capacitor to ground creates high impedance.



- ❖ Wire connection from power supply to op-amp power pin is an inductor.
- ❖ Inductor has high impedance at high frequency.
- ❖ Current draw through op-amp causes current fluctuation in inductor – can feedback to the op-amp.
- ❖ Resulting in op-amp prone to oscillation at high frequency or output overshoot.
- ❖ Add ceramic or tantalum capacitor close to op-amp supply pin to bypass (or decouple) any such high frequency fluctuations.